物理科学学院本科生 2023——2024 学年第 2 学期 热力学与统计物理 小考试卷（一）

学号： 姓名： 成绩：

得 分

一 、（本题 20 分）Suppose $g(U) = C\,U^{3N/2}$, where C is a constant and N is the number of particles. (a) Show that $U = \frac{3}{2}N\tau$. (b) Show that $(\partial^2\sigma/\partial U^2)_N$ is negative.

草稿区

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二 、（本题 20 分）Prove that the probability of finding an atom in any particular energy level is $P[U] = e^{-F/\tau}/Z$, where $F = U - \tau\sigma$ is the free energy.

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三、（本题 25 分）(a) Show that the density of orbitals of free electron in one dimension is $D_1(\varepsilon) = (L/\pi)(2m/\hbar^2\varepsilon)^{1/2}$, where L is the length of the line. (b) Show that in two dimensions, for a square of area A, $D_2(\varepsilon) = Am/\pi\hbar^2$.

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四、（本题 25 分） Please qualitatively draw the variation of a Bose gas' capacity with temperature?

